

International Finance

Lecture I - February 17th

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IES FSV UK – Spring 2026

Moodle: <https://dl1.cuni.cz/course/view.php?id=4605>

(International Finance JEB050/JPB335)

Course enrolment key: **PostUSD?2026**

Outline

- Course organization
- Introduction: What is international finance about?
 - Main features
 - Role of USD/USA
 - Patterns in international finance
- A brief excursion into the history of international finance
 - From safer payments to international lending

Introduction

- Lecturers: Vilém Semerák, Ph.D.
- Teaching assistant: Mgr. Tomáš Karhánek
- E-mails: vilem.semerak@fsv.cuni.cz
47860177@fsv.cuni.cz
- **Class Time:** Tuesday 3:30 p.m. – 4:50 p.m.
- **Office hours:** After class, Tuesdays 5 – 5:30 p.m.
Fridays, 12:30 – 1 p.m.
or by appointment (Zoom/Google Meet possible)
Room 311, IES

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Course Policies

- Registration issues & course capacity
- Semi-mandatory attendance:
 - You need to be present during at least **7 of the 13** sessions to be eligible for a passing grade.
 - Initially, sign-up sheets will be used; a more efficient solution might be used later hopefully.

Grading Policy

- Weekly open-book online quizzes in Moodle: 5%
- Final exam (“on-site” written form only): 75%
 - A **minimum of 45 points** on the final exam is necessary for a passing grade.
- (Team) assignment: 20%
 - Teams with two members
 - Topics and updated rules: March 10th
 - Deadline: May 5th
- Additional bonus points for participation:
 - *up to 5 percentage points, which can be used to replace missing points from assignments or online quizzes.*

Tentative Dates for Final Exams

- Currently planned:

May 26th (**TBC**)

June 2nd

June 9th

June 16th

- Other options can be provided – depending on the situation.

Courses: What is included?

- *International trade*
 - *Standard trade theory*
 - *Trade policy analysis*
- **International finance**
 - Institutional aspects of international finance
 - Balance of payments
 - Exchange rates economics
 - Intro into international macro
- Economics of global business
 - More practical/realistic aspects of international business (but mixed with theory and empirical evidence too)
 - Wider coverage: global value chains, international differences relevant for global business, foreign direct investment
- International Political Economy (IPE)
 - Global and interdisciplinary view: functioning of international institutions, global monetary arrangements

Relevant Types of Questions:

- Why is the USD the main global currency and will it be replaced by Chinese RMB, Euro, or Bitcoin?
- How should the USD respond when new protectionist measures are announced? And how do we explain that it responded differently?
- Are balance of trade disequilibria relevant? Will they correct automatically or not?
- Why are exchange rates so volatile?
- Can we predict exchange rates?
- How does the adoption of a fixed exchange rate regime change the policy options for governments/central banks?
- What is likely to happen in countries that have adopted Euro recently (Bulgaria)?
- Should the Czech Republic adopt the Euro?
- Was Chinese currency really unfairly and deliberately “undervalued”? How can we test this?

Course Outline (Changes Possible)

1. Introduction to international finance. Foreign exchange market: main actors, their role, types of interactions. Brief outline of the history of international finance. (February 17th)
2. International transfers/payments. Channels, related instruments. February (24th)
3. Balance of payments structure and national income accounting for an open economy. International investment position (March 3rd)
4. Introduction to exchange rate determination: asset approach. Covered and uncovered interest rate parity. (March 10th)
5. Money, interest rates and exchange rates. Long-run aspects of exchange rate determination. Purchasing power parity. Overshooting. (March 17th)
6. Prices and exchange rates continued: Fischer effect. Exchange rates and competitiveness: real (effective) exchange rates, Balassa-Samuelson effect. Application on the nominal and real convergence in the EU (March 24th)
7. Introduction to the AA-DD model. Temporary and permanent changes in fiscal and monetary policies. (March 31st)
8. The AA-DD model: XX curve. Application on policy analysis. (April 7th)
9. Policy options for reaching internal and external balance. Swan diagram. Fixed exchange rates: macroeconomic implications. Interventions and sterilisation. Policy trilemma. (April 14th)
10. Monetary integration: costs and benefits. Monetary integration: effects – empirical research. (April 21st)
11. A brief introduction to the models of balance of payments crises. (April 28th)
12. Debts and (developing) countries. **Deadline for the team assignment/paper** (May 5th)
13. The future of IMS, role of the USD. Discussion of current topics. (May 12th)
14. Reserved for a make-up class (possibly for April 21st). (May 19th)

Reading for the Course

- Main text:
 - **Krugman & Obstfeld & Melitz: International Economics** (parts III and IV)
- Alternative options:
 - Pilbeam: International Finance
 - Wang: The Economics of Foreign Exchange and Global Finance.
 - Appleyard & Field: International Economics
 - Pugel: International Economics
- Additional texts:
 - Levi: International Finance
 - Sercu: International Finance: Theory into Practice
- Additional papers and articles will be recommended (and links to them provided via moodle)

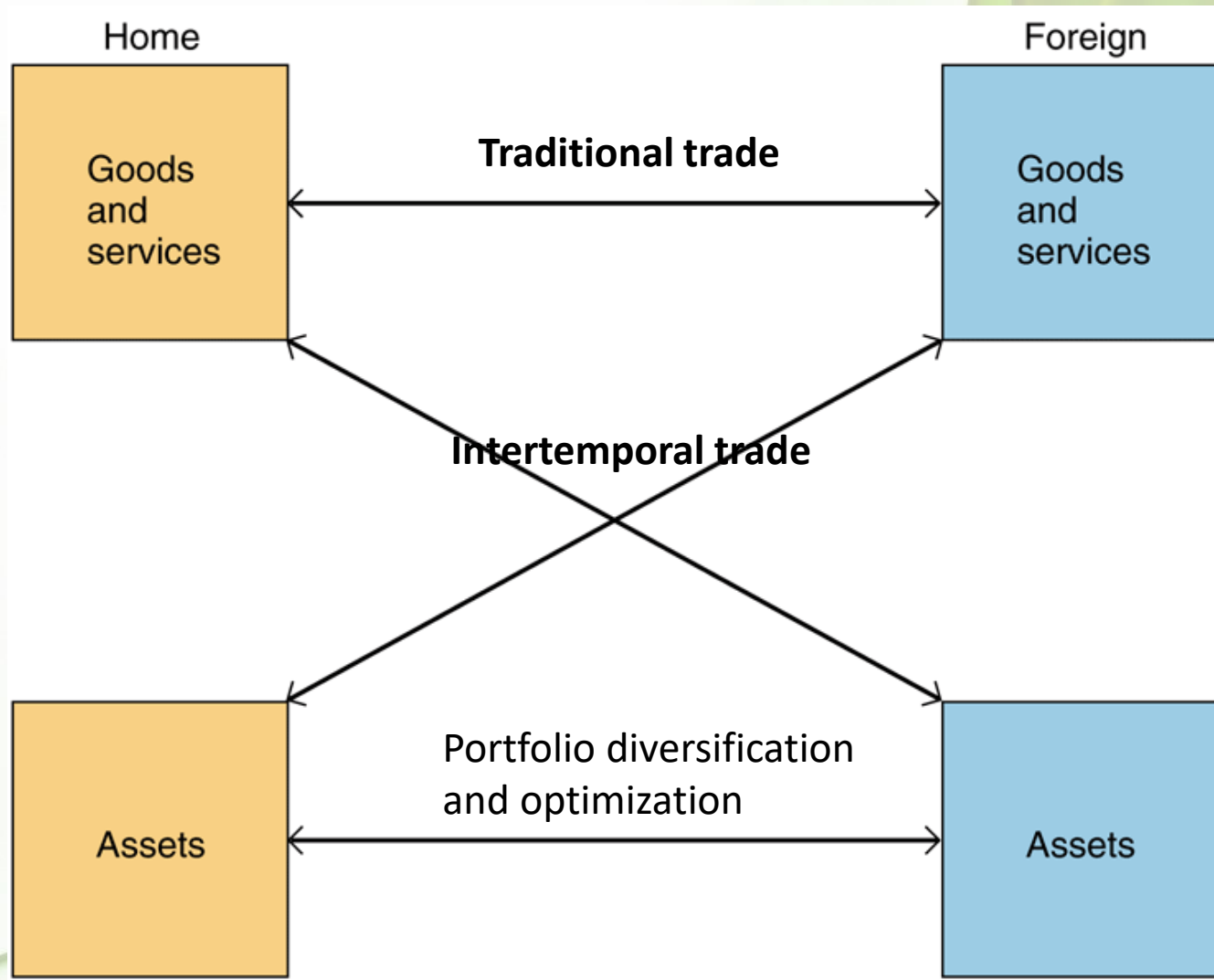
Reading for the Next Week:

- McLeay et. al. (2014): Money in the modern economy: an introduction
 - A simple summary of basic facts and features about money
- King, Osler, Dagfinn (2011) - Foreign Exchange Market Structure, Players and Evolution
 - Section II: Players and Information in FX markets (p. 9-19)
 - Section III: Electronic trading revolution in FX markets (p. 19-37)

The background of the slide features abstract, flowing, and layered green and white lines that create a sense of movement and depth, primarily concentrated on the right side of the frame.

International Finance: Introduction

The Three Types of International Transaction



(International) Finance: Functions

- Basic functions:
 - Mediating payments
 - Providing liquidity
 - Resolving denomination mismatches
 - Reducing credit risk
 - Mediating maturity differences
 - Bearing interest rate
 - Dealing with exchange rate risks
- Who/what can perform them?
 - Institutions (banks)
 - Markets
 - New decentralized platforms

How Much Money?



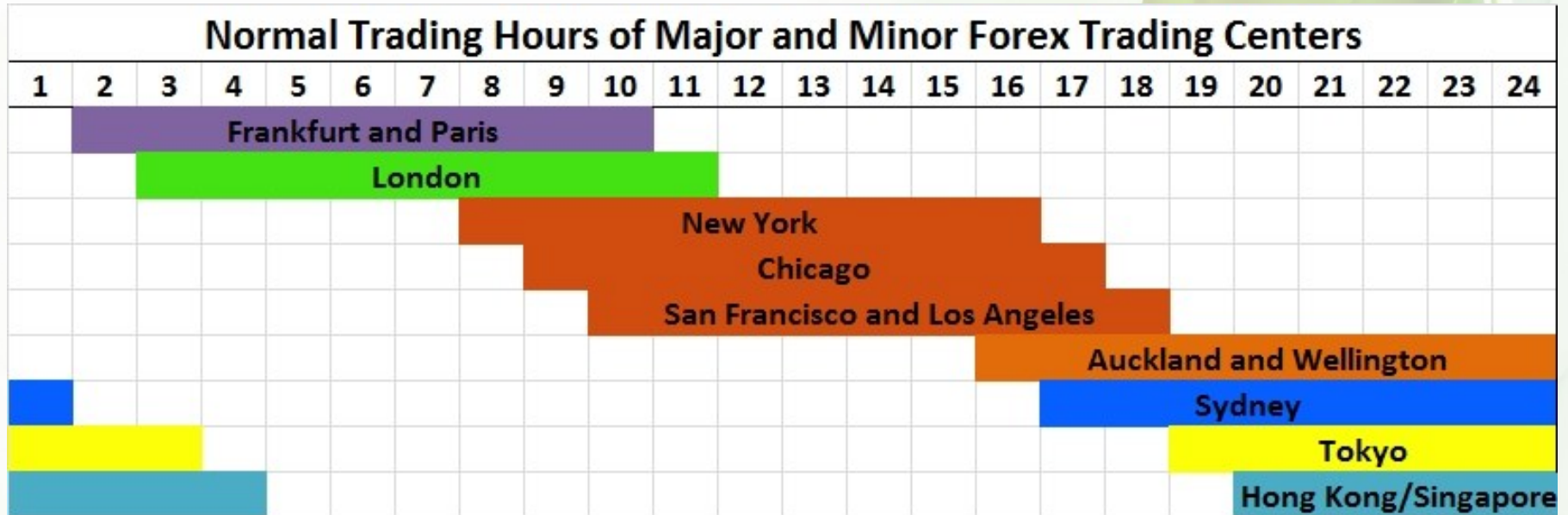
For comparison:

- **annual global trade (merchandise + services) in 2024: 32,2 trillion USD**

Source: BIS Triennial Central Bank Survey, WTO (trade data)

<https://www.bis.org/statistics/rpfx22.htm>

World of Forex Dealing Operates 24-5



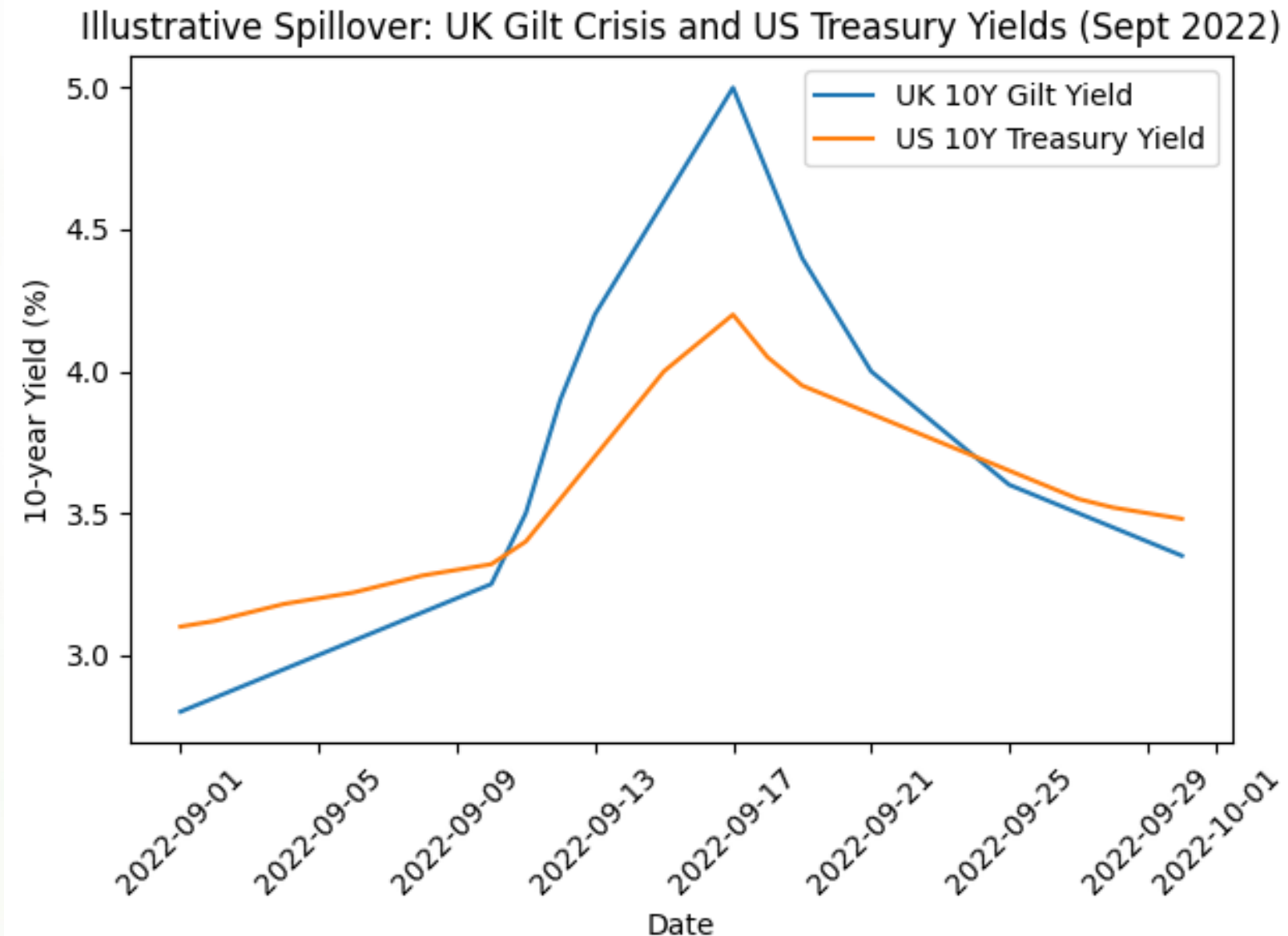
Stylized Facts on Global Economy & Transactions (1)

- Growing interdependence in the global economy → complexity
 - Increasing intensity of trade in goods and services
 - **Even faster increasing intensity of international financial transactions**
 - Much higher volume than necessary for trade flows
 - Increasing **asset interdependence**
 - Best evidence: development during the 2008/9 financial crisis
 - Problems of Cypriot banks (linked to Greek crisis)
 - Increasing importance of emerging markets, but (some) traditional markets still remain rather important

Interdependence Illustration

- In September 2022, the UK government announced a large, unfunded tax-cut package. **Markets panicked.**
- UK government bond (gilt) yields spiked violently — triggering margin calls in pension funds using liability-driven investment (LDI) strategies.
 - UK pension funds were forced to sell assets to raise collateral.
 - That means selling, but they didn't just sell UK gilts.
 - They also sold US Treasuries and other liquid assets.
 - This selling pushed yields even higher. That triggered more margin calls.
- Result: US yields jumped in sympathy with a UK fiscal announcement.

Interdependence Illustration (2)



Stylized Facts on Global Economy & Transactions (2)

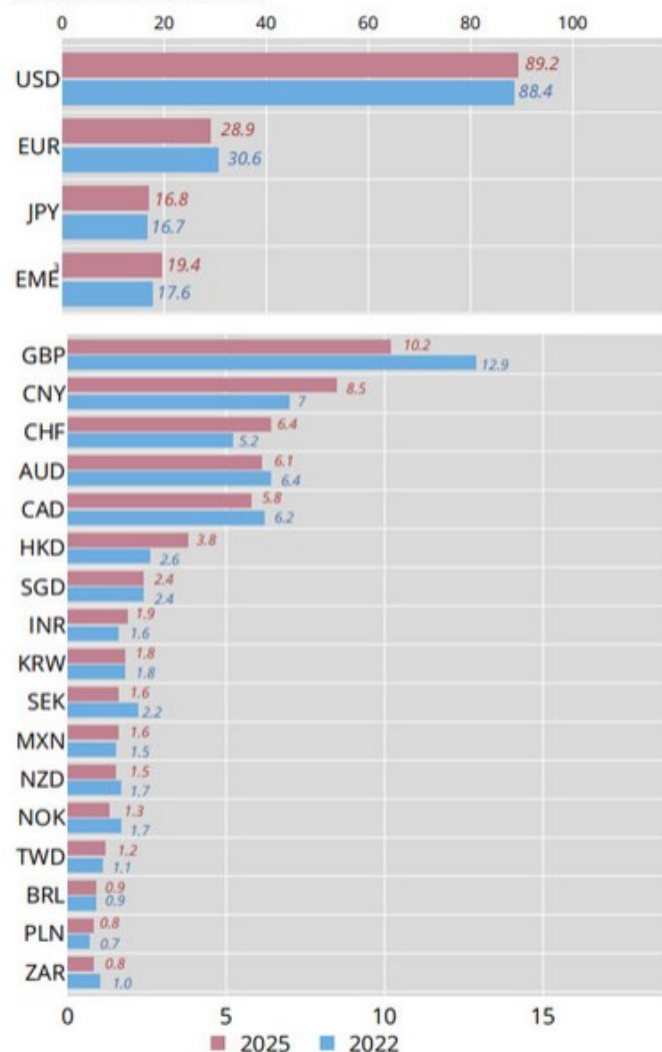
- Financial innovations
 - New types of assets/markets:
 - Cryptocurrencies (not just Bitcoin/altcoins)
 - Stablecoins
 - Various other instruments which can be used as currencies – either for payment or for speculative transactions: electronic trading can make their use more efficient and easier than in the past.
 - Complex derivatives
 - Decentralized transactions
 - E.g. Wise/TransferWise: avoids transaction fees by matching transactions of people in different countries
- Interesting side effects: while huge volumes of data are available, it can still be more difficult to get consistent data and bigger picture

Foreign exchange market turnover by currency and currency pairs¹

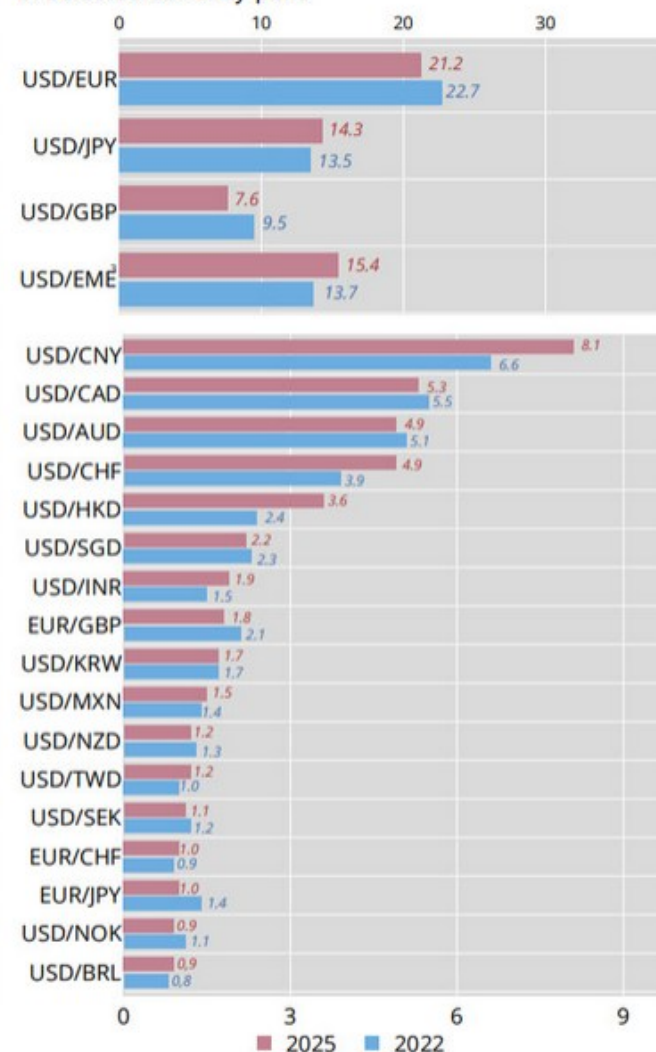
Net-net basis, daily averages in April, as a percentage of total turnover

Graph 2

A. Selected currencies²



B. Selected currency pairs



¹ Adjusted for local and cross-border inter-dealer double-counting, ie "net-net" basis. ² As two currencies are involved in each transaction, the sum of shares in individual currencies will total 200%. ³ Emerging market economy (EME) currencies excluding the Chinese renminbi and Russian rouble: AED, ARS, BGN, BHD, BRL, CLP, COP, CZK, HKD, HUF, IDR, ILS, INR, KRW, MXN, MYR, PEN, PHP, PLN, RON, SAR, SGD, THB, TRY, TWD and ZAR.

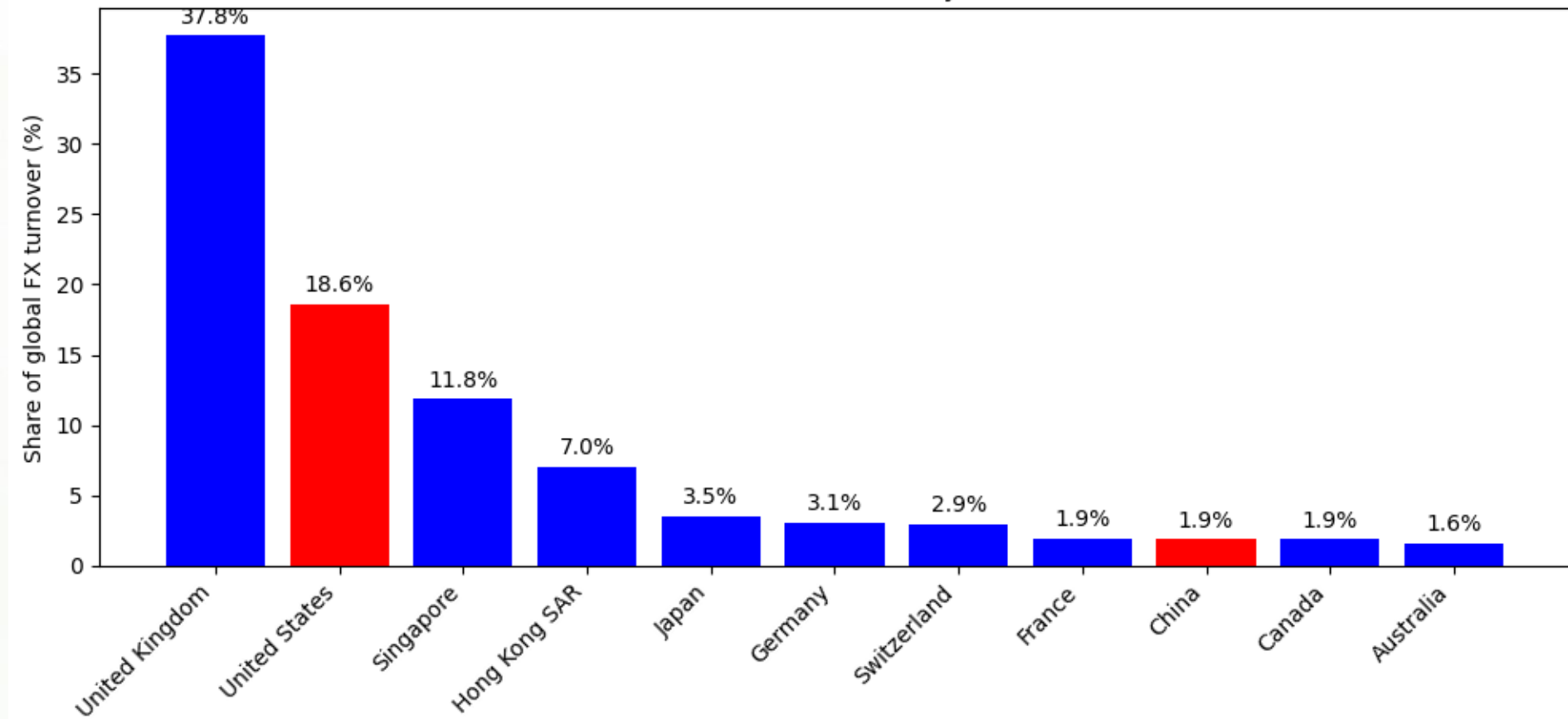
Source: BIS Triennial Central Bank Survey. For additional data by currency and currency pairs, see Tables 4 and 5. See our Data Portal for access to the full set of published data.

Geographical Distribution of Global FOREX Market Turnover (Top 15 = 94.1% in 2022)

Country	2010		2025	
	Amount	%	Amount	%
United Kingdom	1853.6	36.7	4745	37.8
United States	904.4	17.9	2335	18.6
Singapore	266.0	5.3	1485	11.8
Hong Kong SAR	237.6	4.7	883	7.0
Japan	312.3	6.2	440	3.5
Switzerland	249.5	4.9	370	2.9
France	151.6	3.0	242	1.9
Germany	108.6	2.2	386	3.1
Canada	61.9	1.2	243	1.9
China	19.8	0.4	235	1.9
Australia	192.1	3.8	201	1.6
Luxembourg	33.4	0.7	70	1.6
Denmark	120.5	2.4	102	0.9
Netherlands	18.3	0.4	60	0.5
Korea	43.8	0.9	86	0.7

Source of data: BIS

BIS 2025 FX Turnover Shares by Location (%)



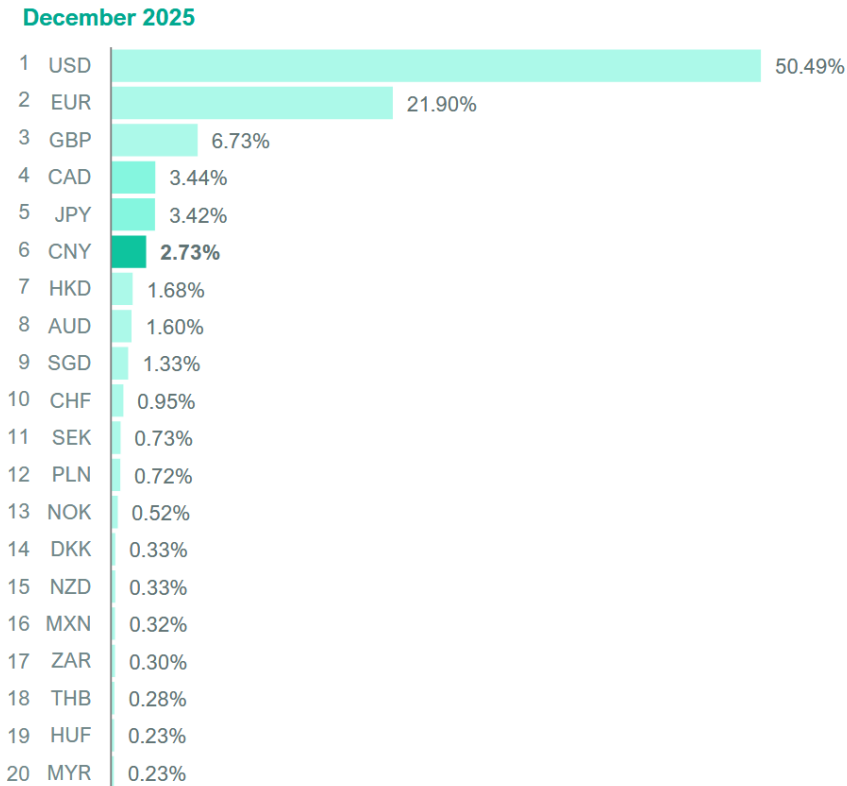
Highlights from the 2025 Triennial Survey of turnover in OTC FX markets:

- Trading in OTC FX markets reached **\$9.6 trillion per day in April 2025** ("net-net" basis, all FX instruments), up 28% from three years earlier.
 - Turnover of FX spot and outright forwards was 42% and 60% higher, respectively. Their shares in global turnover thus increased
 - Turnover of FX options more than doubled.
 - Turnover of FX swaps grew modestly, resulting in a drop in their share to 42%
- **The US dollar continued to dominate global FX markets**, being on one side of 89.2% of all trades.
 - The share of the euro fell to 28.9% (from 30.6%)
 - The share of the Chinese renminbi rose to 8.5%
- **Inter-dealer trading accounted for 46% of global turnover** (almost unchanged from 2022)
 - Just under half of all daily foreign exchange (FX) market transaction volume consisted of large banks and securities dealers trading with each other
 - The share of trading with "other financial institutions" was 50%.
- **Sales desks in the top four jurisdictions – the United Kingdom, the United States, Singapore and Hong Kong SAR – accounted for 75% of total FX trading** ("net-gross" basis).
 - Singapore gained market share, reaching 11.8% of the total.

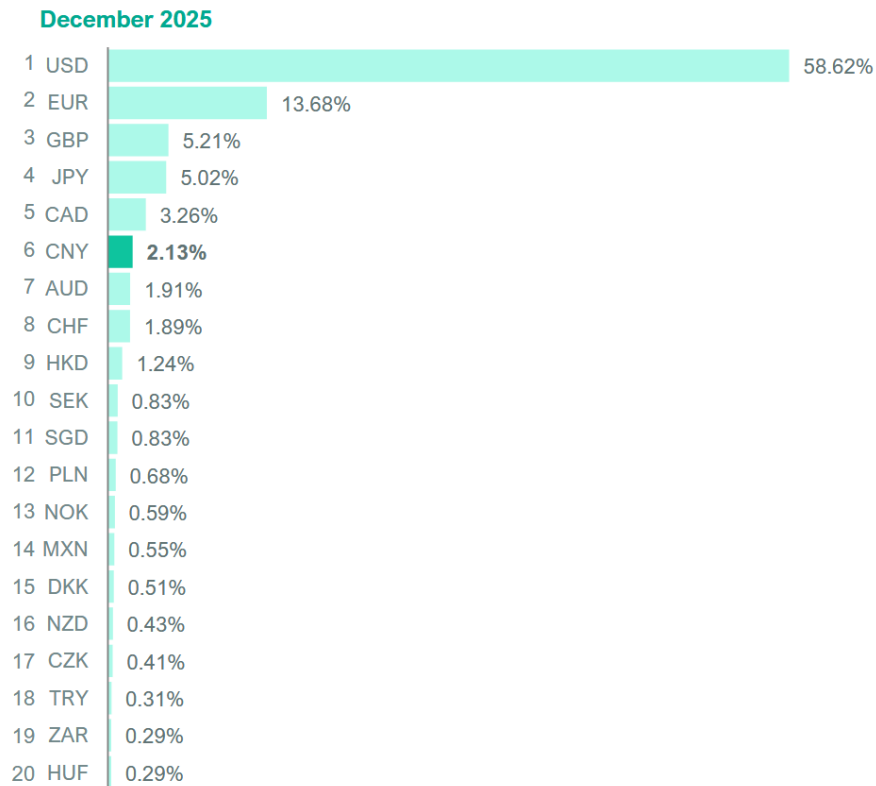
USA/USD Dominates in Finance....

- SWIFT's RMB Tracker (Dec 2025) shows USD about **50.5%** of global payments by value
- **Reserve currency role:** IMF COFER-based reporting shows USD around 58.9% of allocated reserves
- The United States accounts for roughly 60% of global **equity market capitalisation**, while no other single country exceeds ~6%.
- **Global index funds** are US-dominated
- **International portfolio flows** are benchmarked to US indices
- US asset prices heavily influence global risk sentiment
- Dollar-denominated assets anchor global collateral and derivatives markets
- Result: a special position, “**exorbitant privilege**”

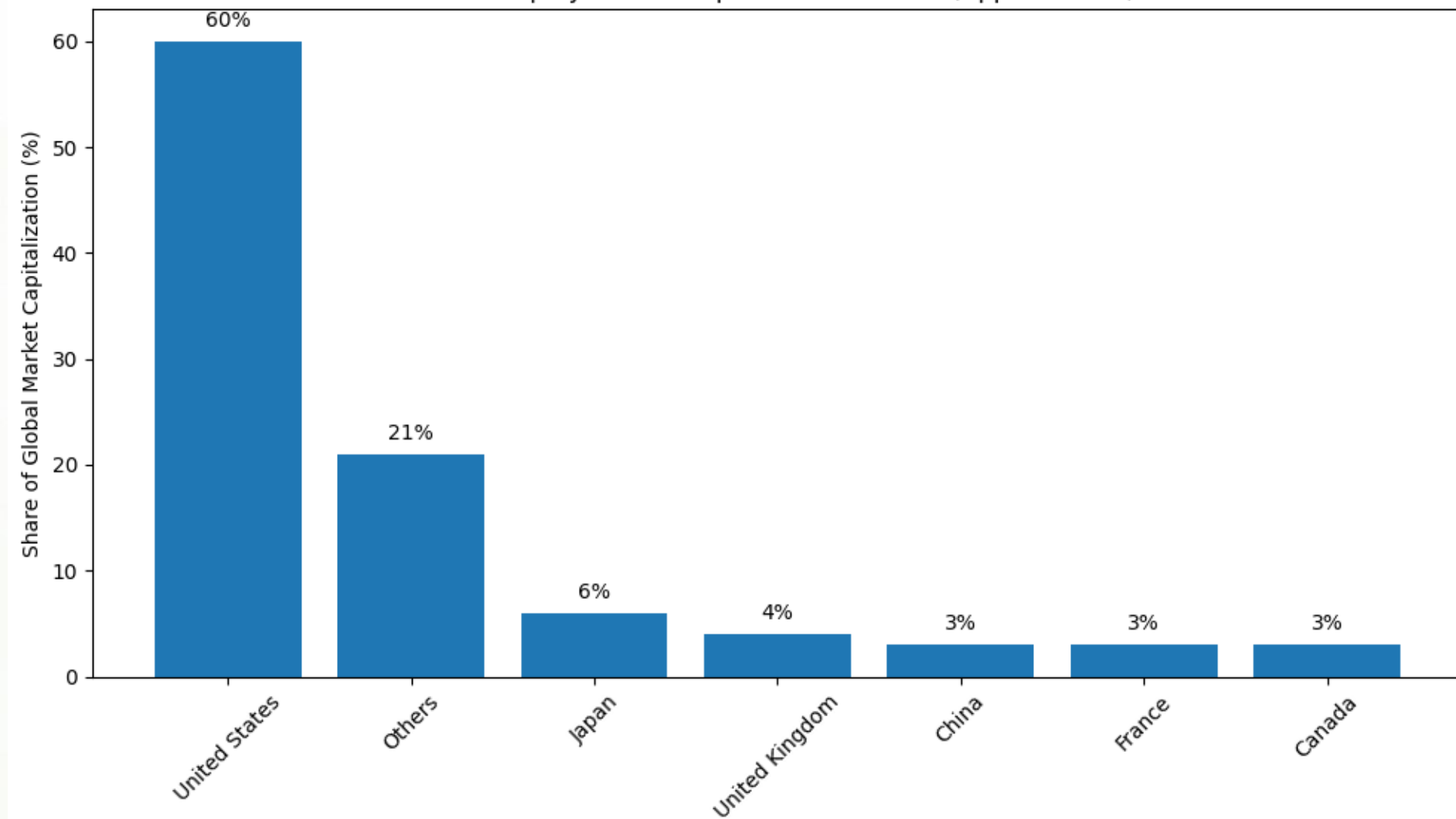
RMB's share as a global payment's currency



RMB's share as an international payment's currency - Excluding payments within Eurozone



Global Equity Market Capitalization Shares (Approx. 2024)

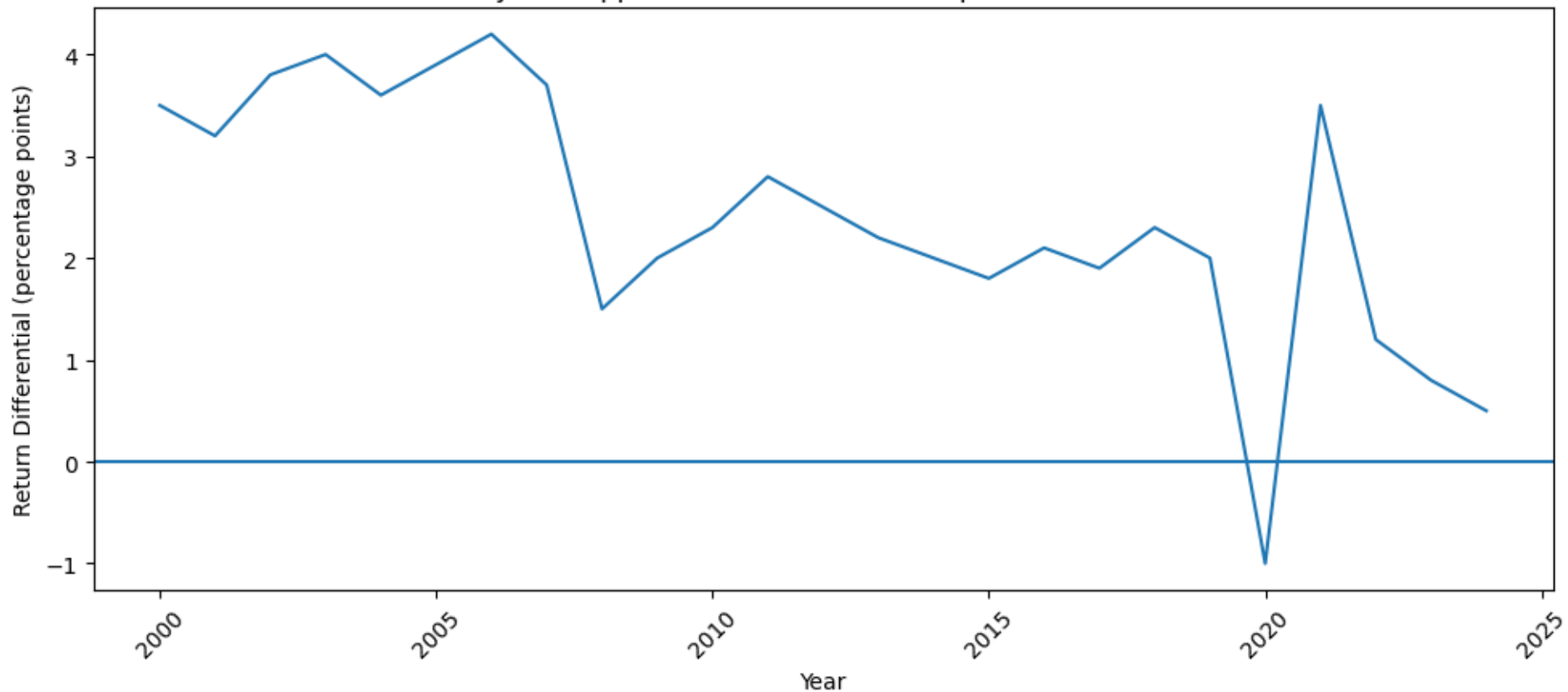


What is the “Exorbitant Privilege”?

- Term which appeared in 1960s (Valéry Giscard d'Estaing?) and described the unique economic benefits the United States derived from the U.S. dollar being the world's primary reserve currency
- What are the benefits?
 - Global investors accept lower yields on USD safe assets and prefer to hold/settle in USD
 - That means lower costs of financing (and hedging) and greater convenience for US borrowers and investors
 - Including the government (investors from other countries tolerate lower yields on US government bonds).
 - Return differentials likely: it is easier for US investors to get higher return on abroad than on liabilities to foreign entities

Approximation (Created with AI Use)

US External Return Differential (Assets minus Liabilities)
Stylized Approximation Based on Empirical Literature



WITH A NEW AFTERWORD

EXORBITANT PRIVILEGE

The Rise and Fall of the Dollar
and the Future of the
International Monetary System

BARRY
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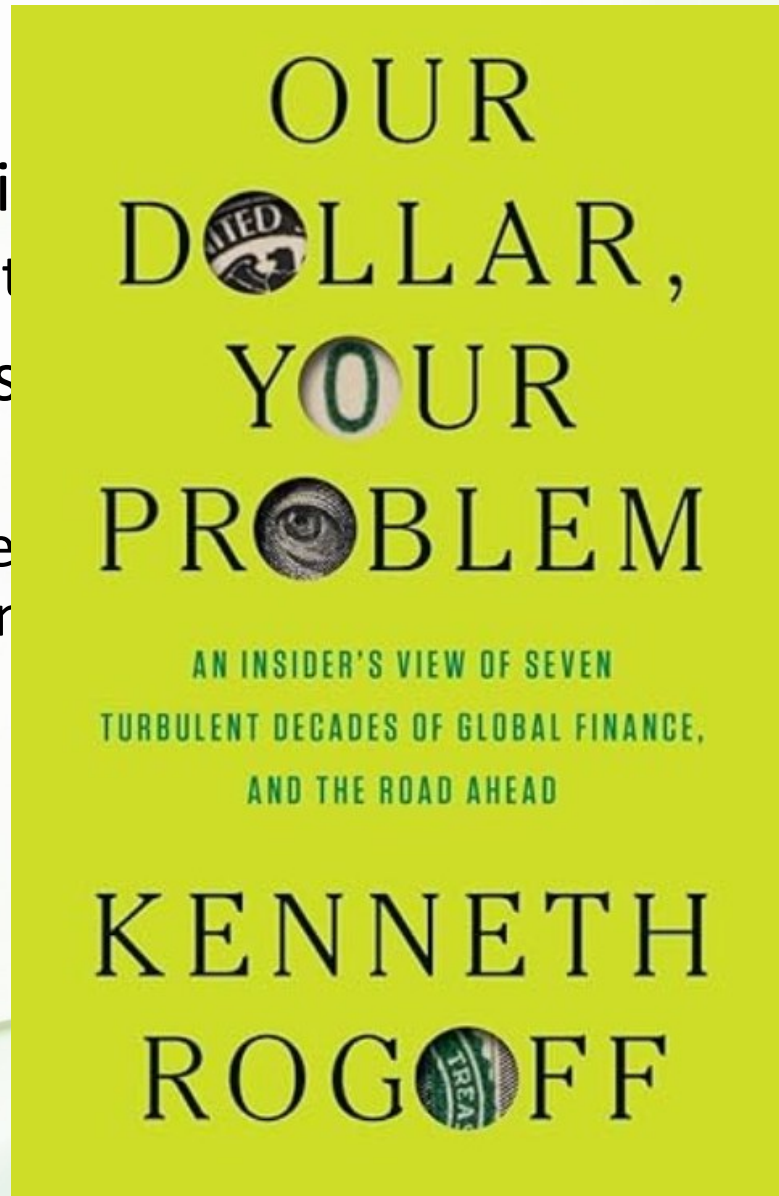
"A fascinating and readable account of
the dollar's rise and potential fall."

—THE ECONOMIST



But USA/USD also undermines the system...

- Reliance on financial markets
 - Motivation to
- Significant loss of confidence by 2025
 - A few interested parties may be willing to move markets to



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Are there any Other Stable Patterns in Global Finance?

- As in the case of trade, the gravity relationship is relevant for flows (e.g. FDIs)
 - But: weaker than in the case of trade
- Strong influence of the market quality on flows:
 - **Lucas paradox:**
 - Capital often flows from poor, capital-scarce countries to rich, capital-abundant countries, contrary to standard economic theory.
- Strong home bias
 - **Equity home bias puzzle:**
 - Individuals and institutions in most countries hold only modest amounts of foreign equity, and tend to strongly favour company stock from their home nation
 - **Feldstein-Horioka puzzle:**
 - Domestic savings and investment rates are highly correlated

Finally: Instruments and Assets

- The range of financial instruments traded international is rather wide, i.e. not only equity, bonds, simple derivatives, gold
 - Rather complex derivatives
 - CDOs - Collateralised Debt Obligations
 - CDS – Credit Default Swap
- And some lesser-known examples:
 - “Cat Bonds” - Catastrophe bonds
 - Insurance for disasters (e.g. hurricanes)
 - Insurance company issues bonds through an investment bank
 - If no disaster happens → coupon paid
 - If a disaster happens → principal forgiven, insurance company has money to pay out claim holders
 - Weather derivatives
 - Contracts paying on temperature averages, heating degree days, rainfall levels
 - Origin: US energy utilities
 - Mortality-linked securities (Death-linked securities)
 - Investors buy insurance policies from elderly individuals and collect when they die.
 - Indeed: international investors have bought a portfolio of US life insurance contracts
 - And of course: cryptocurrencies/assets
 - And do you still remember NFTs (non-fungible tokens)?

A Brief Summary

- The volume of global trading is much bigger than would be needed for narrowly defined international trade transactions
- Persistent role of the USA/USD and of selected traditional centres (London)
 - Alternatives are rising slowly
- Dominant role of the USD and US assets provides an “exorbitant privilege” to the USA
- Global finance is truly global, yet with strong home bias
- There are endless possibilities how to design new types of contracts/assets that can be used in finance
- Turbulent years might be ahead of us

References

- King, Osler, Dagfinn (2011) - Foreign Exchange Market Structure, Players and Evolution
- McLeay et. al. (2014): Money in the modern economy: an introduction
- Melvil & Norrbin: International Money and Finance, chapter 1
- Neal (2015): Concise History of International Finance: Chapters 2 & 3